Group 5 Jack and the Beanstalk Formal presentation

OBJECTIVE

~~Refurbish existing project:~~

~~*Interactive Starlight*~~

Promote reading among children through the use of technology to spark an interest:

Jack and the Beanstalk

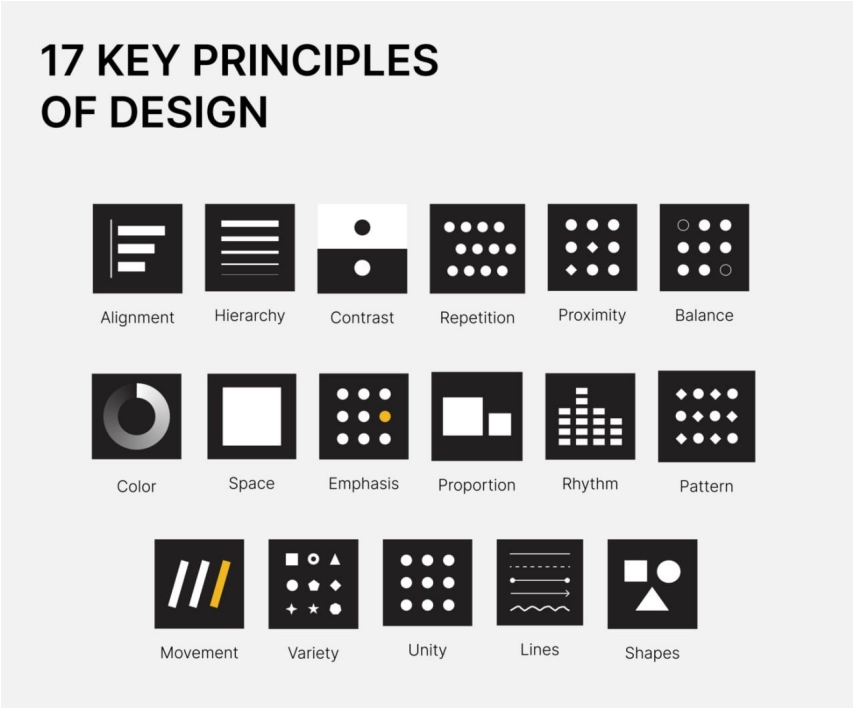
Before

After

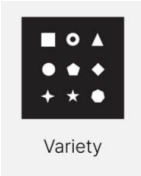
Mood board

- Organize inspirations
- Set the color scheme and tone of the project

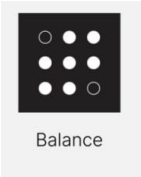
Design principles



Design principles application



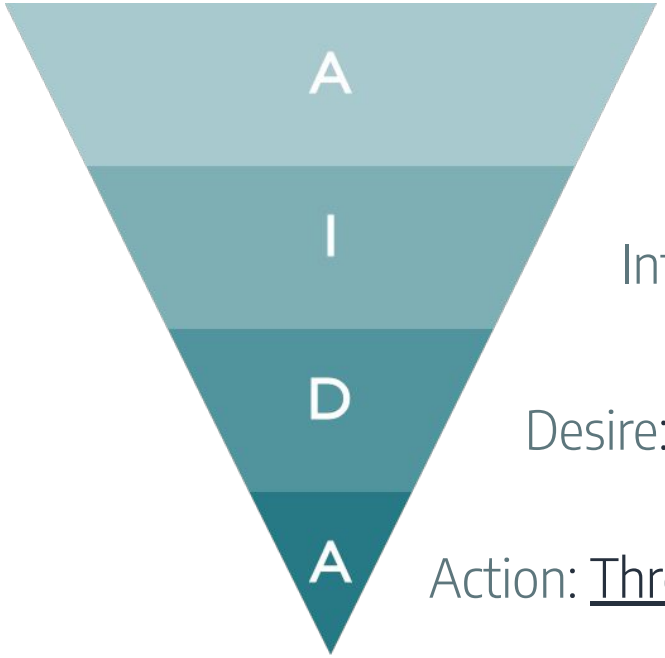
Variety - Lay the decorative items on the base with the goal of putting emphasis on the natural farm environment



Balance - All sides of the showcase are balanced in a way that highlights the difference of each section

*Balance ≠ Symmetry

AIDA Model



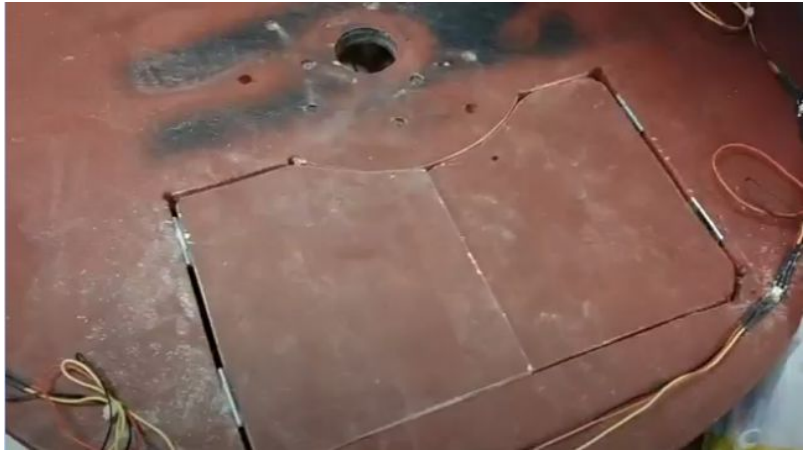
Attention: Loud colors to attract attention

Interest: Interest is built through aesthetic pieces

Desire: Develop desire to view what the workpiece can do

Action: Throw a coin into the well and start the show!

Maintainability & Ease of use



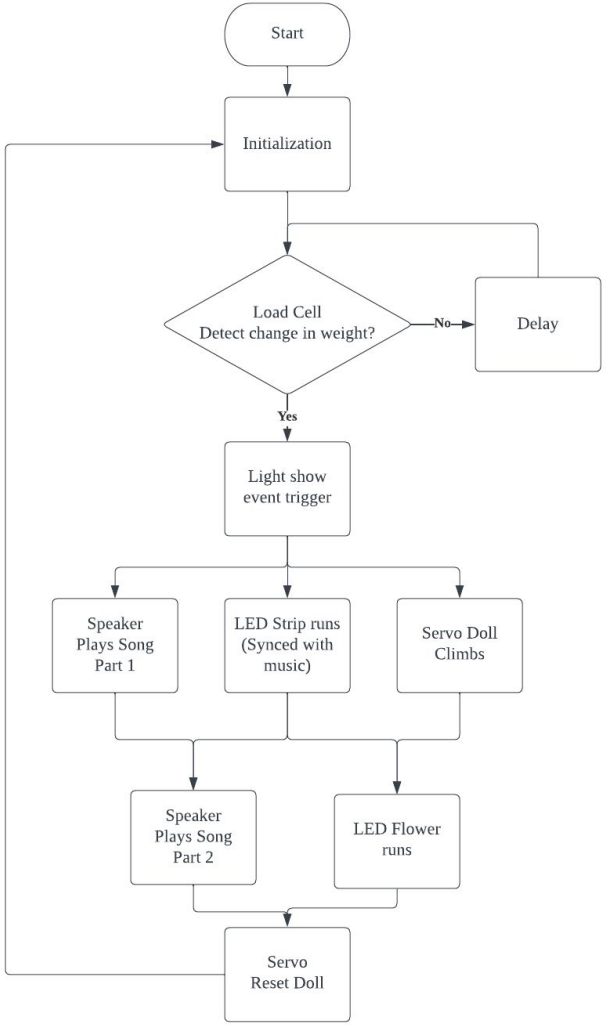
Base compartment access

Roof compartment access

Event trigger

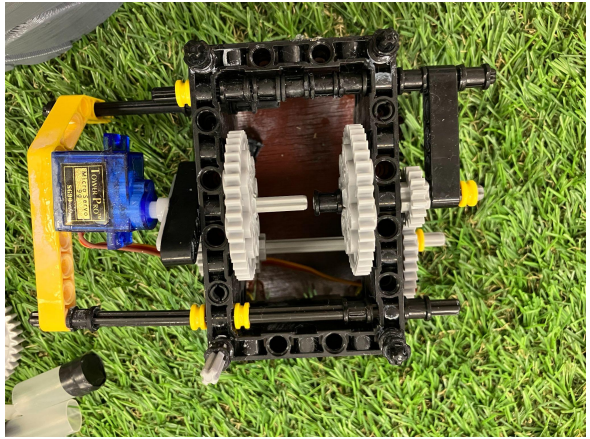
- Custom designed and printed well with coin filter
- Filters out 10 and 20 cent coins
- Only accepts 50 cents and 1 dollar coins

Flowchart



LED Flowers

LED Strip Light Post

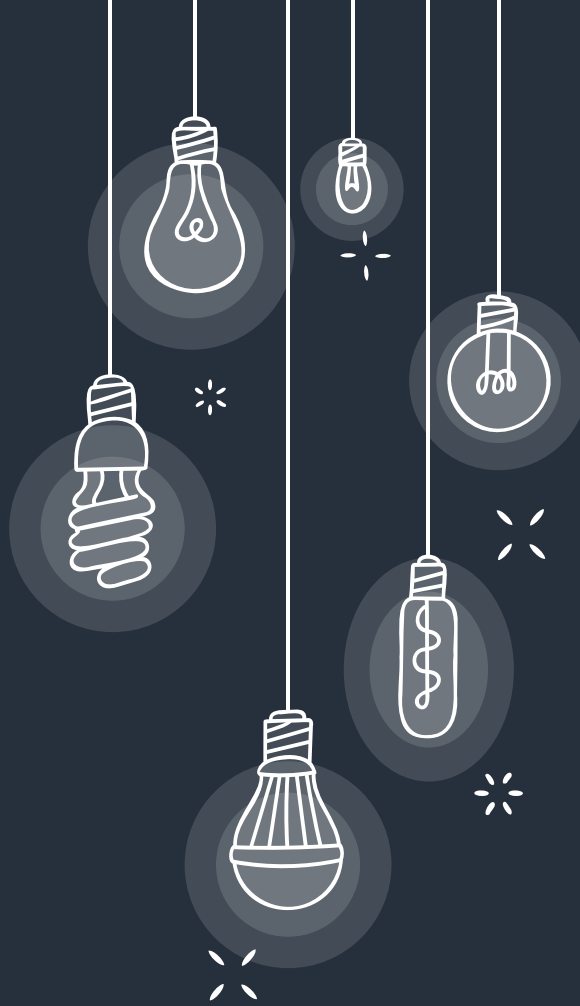


Servo (Actual implementation hidden in housing)

Video

Bill of material

No.	Name	Qty	Cost per item	Cost as per Qty	Link	Remarks	Approved?
				245.15		Budget: <= $300	
				Total (SGD)			
1	100 caterpillar(4rmb)	1	0.8	0.8	https://item.taobao.com/ite		☑
2	beam that hold the gear and(8rmb)	1	1.6	1.6	https://item.taobao.com/ite		☑
3	axle that connect gear(9rmb)	1	1.8	1.8	https://item.taobao.com/ite		☑
4	16teeth gear*10 40 teeth gear *10 (6.9 rmb)	1	1.4	1.4	https://item.taobao.com/ite		☑
5	oversea shipping fee	1	8	8		Combined shipping fee for item 1-4	☑
6	light Clay (White, 50g)	2	3.65	7.3			☑
7	Clay (stone, 1kg)	1	7.5	7.5			☑
8	Cotton Wool					Price and quantity tbc at store. We are currently using those we brought from home, will only purchase if not enough	☑
9	Crepe papers (assorted colours)	2	1	2			☑
10	mesh wire	1	13.9	13.9		Price and quantity tbc at store (for cloud extension support)	☑
11	spray paint	1	5	5			☑
12	acrylic board (White) (60x60cm)	1	51.36	51.36			☑
13	grass patch (50x50cm)	2	6.21	12.42			☑
14	grass patch (30x30cm)	1	2.14	2.14			☑
15	bunny ornament	1	2.14	2.14			☑
16	ice cream stick	1	1.4	1.4			☑
17	Plastic Standoff (50 PCs)	1	13.71	13.71	https://www.lazada.sg/prod		☑
18	8G Glue Sticks (5mm)	2	2.5	5		Hot glue gun glue sticks	☑
19	vanguard sheet/contruction paper	1	1.08	1.08		Price and quantity tbc at store	☑
20	Clay (peach, 500g)	1	4	4			☑
21	Raspberry Pi IO Ribbon	1	9.63	9.63			☑
22	Weight Sensor (5kg) + HX711 Weighing Sen	1	7.49	7.49			☑
23	flower petal	2	1.2	2.4	https://shopee.sg/product/2		☑
24	Astar Leather Adhest (super glue)	1	2.8	2.8			☑
25	acrylic paint	1	7.11	7.11		12c set	☑
26	hinge	2	3	6			☑
27	Pentel Hi Polymer Glue	1	2.4	2.4		Stick glue	☑
28	extension socket	1	15.9	15.9	https://shopee.sg/2-3-4-5-6	5 gang, 3 meter	☑
29	load cell acrylic	1	10.7	10.7			☑
30	Super Glue Twin 2x3ml	1	5.5	5.5			☑
31	Steel bolts & nuts M4x20mm	1	3	3			☑
32	Steel bolts & nuts M8x70mm	1	3	3			☑
33	Half Round L Bracket 40mm	1	2.4	2.4			☑
34	Zip ties (100 pcs)	2	2.4	4.8			☑
35	Velcro	1	2.28	2.28	https://shopee.sg/product/4		☑
36	Wall Adapter Power Supply 9VDC 1A - EU	1	10.26	10.26	https://www.sgbotic.com/in		☑
37	Serial MP3 Music Player Module	1	6.93	6.93	https://shopee.sg/RR-UART		☑

THANKS!

Any questions?